

CWA SCIENCE CLASSES

Concept With Abhishek | Class 10 — Chemistry

Chapter 1: Chemical Reactions and Equations

1. A chemical reaction involves a change in which new substances with new properties are formed.
2. Balancing chemical equations follows the Law of Conservation of Mass — atoms are neither created nor destroyed.
3. Types of Reactions: Combination, Decomposition, Displacement, Double Displacement, Oxidation & Reduction (Redox).
4. Combination reaction: $A + B \rightarrow AB$. Example: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$.
5. Decomposition reaction: $AB \rightarrow A + B$, may be thermal, electrolytic or photolytic.
6. Displacement reaction: More reactive element displaces less reactive element from its compound.
7. Double displacement: exchange of ions between two compounds, often forms a precipitate.
8. Oxidation is gain of oxygen/loss of hydrogen; Reduction is loss of oxygen/gain of hydrogen. Corrosion & rancidity are effects of oxidation in daily life.